

1.(a) charge: area of just

$$A = (4 \times 5 \times 0.5 + 4 \times 10 + 8 \times 6 + 4 \times 8 \times 0.5 + 3 \times 6 \times 0.5) \times 10^3 \\ = (10 + 40 + 48 + 16 + 9) \times 10^3 = \underline{\underline{123 \times 10^3 \text{ C}}}$$

(b) Energy: $W = \int v i dt$. Get Power function.

$$v = \frac{3}{15000}t + 9 = 0.2 \times 10^{-3}t + 9 \quad (0 < t < 15 \text{ ks})$$

$$i = -1.25 \times 10^{-3}t + 15 \quad (0 < t < 4 \text{ ks})$$

$$i = -0.5 \times 10^{-3}t + 12 \quad (4 < t < 12 \text{ ks})$$

$$i = -2 \times 10^{-3}t + 30 \quad (12 < t < 15)$$

$$P \propto \underline{vi}$$

$$0 < t < 4 \text{ ks} ; P = (0.2 \times 10^{-3}t + 9) \cdot (-1.25 \times 10^{-3}t + 15) \\ = -0.25 \times 10^{-6}t^2 - 8.25 \times 10^{-3}t + 135$$

$$W = \int_0^{4000} P dt = 468.667 \text{ kJ}$$

$$4 < t < 12 ; P = -0.1 \times 10^{-6}t^2 - 2.1 \times 10^{-3}t + 108$$

$$W = \int_{4000}^{12000} P dt = 674.133 \text{ kJ}$$

$$12 < t < 15 ; P = -0.4 \times 10^{-6}t^2 - 12 \times 10^{-3}t + 270$$

$$W = \int_{12000}^{15000} P dt = 104.4 \text{ kJ}$$

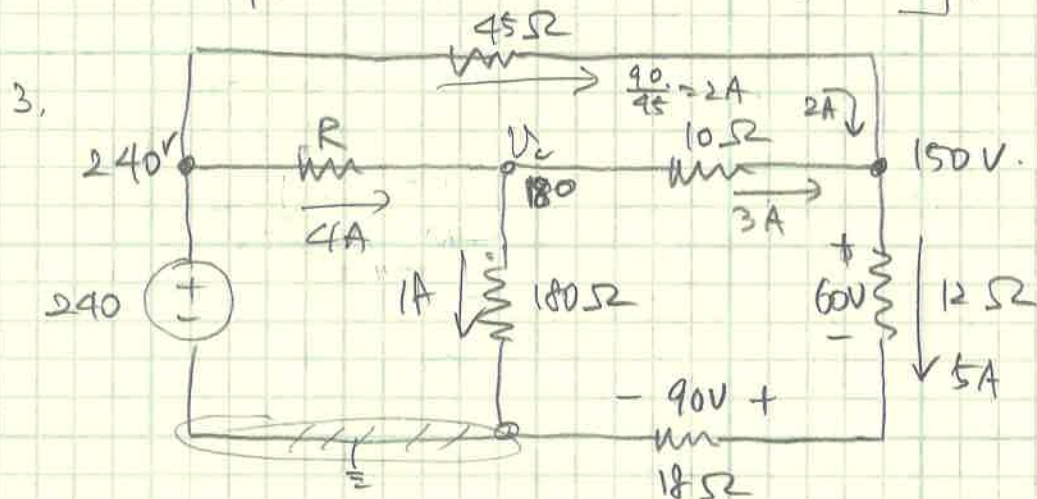
$$\therefore \text{total energy} \propto 1247.2 \text{ kJ}$$

2 (a) left loop K.V.L. $-5 + 54000 i_1 - 1 + 6000 (31 i_1) = 0$
 $-6 + 240000 i_1 = 0 \quad i_1 = 2.5 \times 10^{-5} \text{ A} = \underline{25 \mu\text{A}}$

(b) Big loop $-5 + 54000 \times i_1 - 1 + V - 1800 \times 30 \times i_1 + 8 = 0$
 $2 + 54000 \times 25 \times 10^{-6} + V - 54000 \times i_1 = 0$
 $\underline{V = -2 \text{ V}}$

(c) $5 \text{ V} : 5 \times 25 \times 10^{-6} \quad \text{supply} \quad 125 \times 10^{-6}$
 $1 \text{ V} : 1 \times 25 \times 10^{-6} \quad \text{supply} \quad 25 \times 10^{-6}$
 $30 i_1 : 2 \times 30 \times 25 \times 10^{-6} \quad \text{absorb}$
 $8 \text{ V} : 8 \times 30 \times 25 \times 10^{-6} = 6000 \times 10^{-6} \quad \text{supply}$
 $\therefore (6000 + 125 + 25) \times 10^{-6} \quad \text{supply} \quad \boxed{6150 \mu\text{W}}$

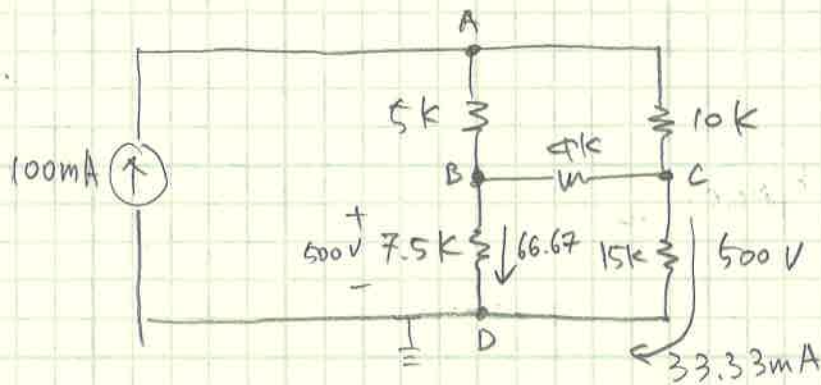
(d) $54000 \times (25 \times 10^{-6})^2 = 33.75 \mu\text{W}$
 $1800 \times (30 \times 25 \times 10^{-6})^2 = 1012.5 \mu\text{W}$
 $6000 \times (31 \times 25 \times 10^{-6})^2 = 3603.75 \mu\text{W}$
 $30 i_1 : 2 \times 30 \times 25 \times 10^{-6} = 1500 \mu\text{W} \quad] + 6150 \mu\text{W}$



$$V_c - 150 = 30 \quad V_c = 180$$

$$(240 - 180) / 4 = 60 / 4 = \underline{15 \Omega}$$

4.



$$I_{BD} + I_{CD} = 100 \text{ mA}$$

$$I_{BC} = 100 - 33.33 = 66.67 \text{ mA}$$

$$V_B = V_C \rightarrow I_{BC} = 0.$$

$$\therefore \begin{cases} P_{5k} = (66.67 \text{ mA} \cdot 5k\Omega) \cdot 66.67 \text{ mA} = 22.22 \text{ W} \end{cases}$$

$$\begin{cases} P_{10k} = (33.33 \text{ mA} \cdot 10k\Omega) \cdot 33.33 \text{ mA} = 11.10 \text{ W} \end{cases}$$

$$P_{4k} = 0. \quad P_{15k} = 500V \times 33.33 \text{ mA} = 16.67 \text{ W}.$$

$$P_{7.5k} = 500 \times 66.67 = 33.33 \text{ W}.$$